NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Course Name: ETHICAL HACKING

Assignment- Week 7

TYPE OF QUESTION: MCQ/MSQ/SA

Number of questions: 10 **Total mark: 10 x 1 = 10**

QUESTION 1:

A message M is fed to a hash function to generate hash value H = HASH(M). Which of the following statement(s) is/are **true**?

a. The number of bits in H is typically much smaller than the number of bits in M.
b. The number of bits in H and M are always exactly equal.
c. Hashing produces a fixed-length output regardless of input size.
d. The output length of a hash function grows proportionally with the input length.

Correct Answer: a, c

Detailed Solution: A hash function compresses an arbitrarily large input into a small, fixed-length output, regardless of input size. In hashing output length does not grow with input length.
Thus, the correct options are (a) and (c).

QUESTION 2:

Which of the following statement(s) is/are **true**?

a. The block size of DES is 64 bits.
b. The key size of DES is 128 bits.
c. The hash digest size of MD5 is 128 bits.
d. SHA-256 produces a 512-bit digest.
e. Triple DES uses a fixed 512-bit key.

Correct Answer: a, c

Detailed Solution: DES uses a 64-bit block size (with a 56-bit key, not 128 bits). MD5 produces a 128-bit digest, SHA-256 produces a 256-bit digest (not 512), and Triple DES's effective key length is up to 168 bits, not 512.

Thus, the correct options are (a) and (c).

QUESTION 3:

Which of the following statement(s) is/are **true** regarding SSL/TLS protocols?

a. HTTPS relies on SSL/TLS to secure HTTP communication.
b. SSL/TLS guarantees faster data transmission compared to plain HTTP.
c. TLS operates at the physical layer of the network stack.
d. The SSL Record Protocol guarantees non-repudiation of the sender.

Correct Answer: a

Detailed Solution: SSL/TLS (and HTTPS, built on it) operate above the IP layer, and the SSL Record Protocol handles data encryption. SSL/TLS adds cryptographic overhead rather than guaranteeing faster transmission; it does not operate at the physical layer, and the Record protocol does not provide non-repudiation.

Thus, the correct option is (a).

QUESTION 4:

Which of the following statement(s) is/are **true** regarding collision resistance in hash functions?

a. It is a weaker requirement than second preimage resistance.
b. It should be computationally difficult to find any two distinct messages producing the same hash.
c. Collision resistance guarantees encryption confidentiality.
d. Collision resistance is trivial to achieve for any hash function design.
e. Collision resistance applies only to keyed hash functions (MACs).

Correct Answer: b

Detailed Solution: Collision resistance requires that finding any two distinct messages with the same hash is computationally infeasible, generally a stronger (or equally strong) requirement than second preimage resistance. Collision resistance is unrelated to confidentiality, is not trivial to achieve, and applies to hash functions generally, not just keyed ones.

Thus, the correct option is (b).

QUESTION 5:

Which of the following statement(s) is/are **true** regarding digital signature verification?

a. The verifier uses the signer's public key to verify a signature.
b. The verifier uses the signer's private key to verify a signature.
c. Successful verification confirms the message has not been altered and was signed by the claimed signer.
d. Verification requires the signer to be online and actively participating.

Correct Answer: a, c

Detailed Solution: Verification uses the signer's public key, which anyone can access, allowing confirmation of both message integrity and signer identity. It does not require the signer's active participation.
Thus, the correct options are (a) and (c).

QUESTION 6:

Which property of digital signatures ensures that the original signer cannot later deny having signed a message?

a. Confidentiality
b. Integrity
c. Non-repudiation
d. Availability
e. Authentication

Correct Answer: c

Detailed Solution: Non-repudiation ensures that once a message is signed with the signer's private key, the signer cannot deny having signed it, because anyone can verify the signature with the corresponding public key.

Thus, the correct option is (c).

QUESTION 7:

Which of the following statements is **false** regarding keyed vs unkeyed hash functions?

a. An unkeyed hash function can be used to verify data integrity when the hash value is obtained from a trusted source.
b. A keyed hash function requires a secret key as part of its computation.
c. An unkeyed hash function inherently provides authenticity.
d. HMAC can be used to provide both integrity and a degree of authentication.

Correct Answer: c

Detailed Solution: Unkeyed hash functions like MD5/SHA-256 provide integrity but not authenticity, since anyone can compute the hash without needing a secret key.

Thus, the correct (false) option is (c).

QUESTION 8:

In IPSec, which mode encrypts/protects only the payload (transport layer data) while keeping the original IP header intact?

a. Tunnel mode
b. Transport mode
c. Replay mode
d. Confidential mode
e. None of these

Correct Answer: b

Detailed Solution: IPSec Transport mode protects only the payload (transport layer data), leaving the original IP header unencrypted/intact, whereas Tunnel mode encapsulates and protects the entire original IP packet, including its header. "Replay mode" and "Confidential mode" are not standard IPSec terms.

Thus, the correct option is (b).

QUESTION 9:

Hash digest size produced by the MD5 algorithm is _____ bits.

Correct Answer: 128

Detailed Solution: MD5 always produces a fixed-length hash digest of 128 bits, regardless of the size of the input message.

Thus, the correct answer is 128.

QUESTION 10:

The block size used by the SHA-512 hashing algorithm is _____ bits.

Correct Answer: 1024

Detailed Solution: SHA-512 processes the input message in blocks of 1024 bits, producing a 512-bit hash digest.
Thus, the correct answer is 1024.

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